

10 ל"ג

7 י"ד

ה'תש"ס ח' שבט ה'תש"ס

(67)

$$\int \sin(5x) \cdot \cos x \, dx =$$

15/10/1

$$\sin x \cdot \cos y = \frac{1}{2} (\sin(x+y) + \sin(x-y))$$

$$\frac{1}{2} \int \sin(5x+x) + \sin(5x-x) \, dx$$

$$= \frac{1}{2} \int \sin(6x) + \sin(4x) \, dx$$

$$\frac{1}{2} \left[-\frac{\cos(6x)}{6} - \frac{\cos(4x)}{4} \right] + C$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

הצגה
③

$$\cos^2 x + \sin^2 x = 1 \quad \text{①}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

→ חצי

$$\int \sin^2 x, \cos^2 x$$

① הוכחה נמצאת

→ חצי

$$\int \sin^2 x, \cos^2 x$$

הוכחה נמצאת
③

הוכחה

$$\int \sin^3 x \, dx = \int \sin x \sin^2 x \, dx$$

$$= \int \sin x (1 - \cos^2 x) \, dx$$

$$\begin{aligned} z &= \cos x \\ dz &= -\sin x \, dx \end{aligned}$$

$$= - \int (1 - t^2) \, dt = \int t^2 - 1 \, dt =$$

$$\frac{t^3}{3} - t + C$$

$$= \frac{\cos^3 x}{3} - \cos x + C$$

$$\int \cos^5 x \, dx = \int \cos x \cdot \cos^4 x \, dx$$

$$= \int \cos x \cdot (\cos^2 x)^2 \, dx =$$

$$\int \cos x (1 - \sin^2 x)^2 \, dx$$

$$t = \sin x$$

$$dt = \cos x \, dx$$

$$\int dt (1 - t^2)^2 =$$

$$\int (1 - 2t^2 + t^4) \, dt =$$

$$t - \frac{2t^3}{3} + \frac{t^5}{5} + C$$

$$= \sin x - \frac{2}{3} (\sin x)^3 + \frac{(\sin x)^5}{5} + C$$

$$\int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx$$

$$= \frac{1}{2} \left[x - \frac{\sin(2x)}{2} \right] + C$$

$$\int \cos^4 x \, dx = \int (\cos^2 x)^2 \, dx$$

$$= \int \left(\frac{1 + \cos(2x)}{2} \right)^2 \, dx$$

$$\frac{1}{4} \int 1 + 2\cos(2x) + \cos^2(2x) \, dx$$

$$\frac{1}{4} \int 1 + 2\cos(2x) + \frac{1 + \cos(4x)}{2} \, dx$$

$$= \frac{1}{4} \left[x + \frac{2\sin(2x)}{2} + \frac{1}{2}x + \frac{\cos(4x)}{2 \cdot 4} \right] + C$$

— יסודות 37

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$x = 2 \arctan t$$

$$t = \tan \frac{x}{2}$$



$$dx = \frac{2}{1+t^2} dt$$

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$$\int \frac{dx}{4 \sin x + 3 \cos x + 5} =$$

$$\int \frac{\frac{2}{1+t^2} dt}{4 \cdot \frac{2t}{1+t^2} + 3 \cdot \left(\frac{1-t^2}{1+t^2} \right) + 5} =$$

$$= \int \left(\frac{2}{1+t^2} + \frac{7t + \frac{3-3t^2}{1+t^2} + 5+5t^2}{1+t^2} \right) dt$$

$$= \int \frac{2}{2t^2 + 7t + 3} dt$$

$$= \int \frac{1}{t^2 + 4t + 4} dt$$

$$= \int \frac{1}{(t+2)^2} dt$$

$$\int (t+2)^{-2} dt =$$

$$\frac{(t+2)^{-1}}{-1} + C =$$

$$-\frac{1}{t+2} + C$$

$$= -\frac{1}{\tan\left(\frac{x}{2}\right) + 2} + C$$

(7)

$$\int \frac{\cos x}{1 + \cos x} dx$$

$$\int \frac{\frac{1-t^2}{1+t^2}}{1 + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$\int \left(\frac{1-t^2}{1+t^2} + \frac{1-t^2}{1+t^2} \right) \cdot \frac{2}{1+t^2} dt$$

$$\int \frac{(1-t^2)(\cancel{1+t^2})}{(1+t^2)\cancel{2}} \cdot \frac{\cancel{2}}{\cancel{1+t^2}} dt$$

$$\int \frac{1-t^2}{1+t^2} dt$$

נפרק לגורמים:

$$\frac{\overbrace{-t^2 + 1}^{-1}}{-t^2 - 1}$$

$$= \int -1 + \frac{2}{t^2 + 1} dt$$

$$= -t + 2 \arctan t + C$$

$$= -\tan \frac{x}{2} + 2 \text{ or } (\tan (\tan \frac{x}{2}) + C$$

$$= -\tan \frac{x}{2} + 2 \left(\frac{x}{2} \right) + C$$

Example 1

(8)

$$\int \frac{\sqrt[6]{x}}{1 + \sqrt[3]{x}} dx = \left| \begin{array}{l} t^6 = x \rightarrow t = \sqrt[6]{x} \\ 6t^5 dt = dx \end{array} \right.$$

$$\int \frac{\sqrt[6]{t^6}}{1 + \sqrt[3]{t^6}} \cdot 6t^5 dt$$

$$= \int \frac{t}{1 + t^2} \cdot 6t^5 dt$$

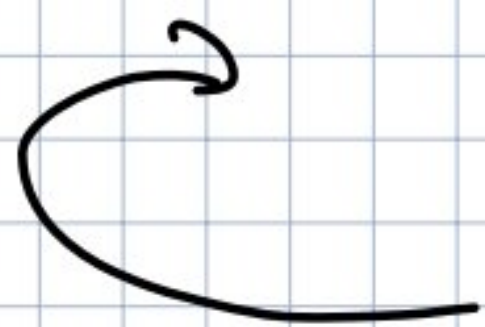
$$= \int \frac{6t^6}{t^2+1} dt$$

$$= 6 \int \frac{t^6}{t^2+1} dt$$

$$\begin{array}{r}
 t^4 - t^2 + 1 \\
 \hline
 t^6 \overline{) t^2 + 1} \\
 - t^6 + t^4 \\
 \hline
 -t^4 \\
 -t^4 - t^2 \\
 \hline
 -t^2 \\
 -t^2 + 1 \\
 \hline
 -1
 \end{array}$$

$$\int t^4 - t^2 + 1 + \frac{-1}{t^2+1} dt$$

$$= \frac{t^5}{5} - \frac{t^3}{3} + t - \arctan t + C$$



$$t = \sqrt[6]{x}$$

$$x = t^6$$

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$$\int \frac{7x + 16}{\sqrt[3]{(7x+8)^2} + 2\sqrt[3]{7x+8}} dx$$

2/3

1/3

$$t^3 = 7x + 8$$



$$t = \sqrt[3]{7x+8}$$

$$3t^2 dt = 7 dx$$

$$\frac{3t^2}{7} dt = dx$$

$$\int \frac{7x+8}{\sqrt[3]{(7x+8)^2 + 2\sqrt[3]{7x+8}}} dx$$

$$\int \frac{t^3+8}{\sqrt[3]{(t^3)^2 + 2\sqrt[3]{t^3}}} \cdot \frac{3t^2}{7} dt$$

$$\int \frac{t^3+8}{t^2 + 2t} \cdot \frac{3t^2}{7} dt$$

$$\int \frac{(t^3+8)t}{t^2+2} dt$$

$$\begin{array}{r} t^3 - 2t^2 + 4t \\ \hline t^4 + 2t \end{array} \bigg| t + 2$$

$$- t^4 + 2t^3$$

$$- 2t^3 + 2t$$

$$- 2t^3 - 4t^2$$

$$4t^2 + 2t$$

$$- 4t^2 + 2t$$

$$0$$

$$\frac{3}{2} \int (t^3 - 2t^2 + 4t) dt$$

$$\frac{3}{2} \left[\frac{t^4}{4} - \frac{2t^3}{3} + \frac{4t^2}{2} \right] + C$$



$$t = \sqrt[3]{3x+1}$$

$$\underline{t^3 = 3x+1}$$

$$\int \frac{x+1}{\sqrt[3]{3x+1}} dx$$

$$t^3 = 3x+1 \rightarrow t = \sqrt[3]{3x+1}$$

$$3t^2 dt = 3 dx$$

$$t^3 - 1 = 3x$$

$$\frac{t^3 - 1}{3} = x$$

$$\int \frac{\frac{t^3 - 1}{3} + 1}{\sqrt[3]{t^3}} \cdot t^2 dt$$

$$\int \frac{\frac{t^3 - 1 + 3}{3}}{t} \cdot t^2 dt$$

$$\frac{1}{3} \int (t^3 + 2) t \, dt$$

$$= \frac{1}{3} \int t^4 + 2t \, dt =$$

$$\frac{1}{3} \left[\frac{t^5}{5} + \frac{2t^2}{2} \right] + C$$

$$t = \sqrt[3]{3x+1}$$

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1

7780 7780 ✓

$$\int \cot^6 x \, dx = \int \frac{1}{\tan^6 x} \, dx$$

(1) (2)

2 fro)

7780 7780

(6 7780)

$$t = \tan x$$

$$\arctan t = x$$

$$\frac{1}{1+t^2} dt = dx$$

$$= \int \frac{1}{t^6} \cdot \frac{1}{(1+t^2)} \cdot dt =$$

$$\int \frac{1}{t^6 (t^2+1)} dt$$

$$\frac{1}{t^6 (t^2+1)} = \frac{A}{t} + \frac{B}{t^2} + \frac{C}{t^3} + \frac{D}{t^4} + \frac{E}{t^5} + \frac{F}{t^6} + \frac{Gt+H}{t^2+1}$$

$$1 = A \cdot t^5 (t^2+1) + B \cdot t^4 (t^2+1) + C \cdot t^3 (t^2+1) + D \cdot t^2 (t^2+1) + E \cdot t (t^2+1) + F (t^2+1) + (Gt+H)t^6$$

$$\underline{t=0:}$$

$$1 = F (0^2+1)$$

$$\boxed{F=1}$$

$$1 = A \cdot t^5 (t^2+1) + B \cdot t^4 (t^2+1) + C \cdot t^3 (t^2+1) + D \cdot t^2 (t^2+1) + E \cdot t (t^2+1) + 1 \cdot (t^2+1) + (Gt+H)t^6$$

$$1 = \underline{A}t^3 + \underline{A}t^5 + \underline{B}t^6 + \underline{B}t^4 + \underline{C}t^5 + \underline{C}t^3 \\ + \underline{D}t^4 + \underline{D}t^2 + \underline{E}t^3 + \underline{E}t + \underline{F}t^2 + \underline{G}t^3 + \underline{H}t^6$$

$x(t)$

t :

$$0 = E$$

t^2 :

$$0 = D + 1$$

$\rightarrow$

$$D = -1$$

t^3 :

$$0 = C + E \rightarrow$$

$$C = 0$$

t^4 :

$$B + D = 0 \rightarrow$$

$$B = 1$$

t^5 :

$$A + C = 0 \rightarrow$$

$$A = 0$$

t^6 :

$$B + H = 0 \rightarrow$$

$$H = -1$$

t^7 :

$$A + G = 0 \rightarrow$$

$$G = 0$$

$$= \int \frac{1}{t^2} + \frac{-1}{t^4} + \frac{1}{t^6} + \frac{-1}{t^2+1} dt$$

↙
-t⁻⁴

$$= \frac{t^{-1}}{-1} + \frac{t^{-3}}{3} - \frac{t^{-5}}{5} - \arctan t + C$$

$$= -\frac{1}{t} + \frac{1}{3t^3} - \frac{1}{5t^5} - \arctan t + C$$

↑
t = tan x

! 3 א' 2

$$\int \frac{dx}{\cos^3(x)} =$$

I

(2)

(4 נק' 2)

$$u = \frac{1}{\cos(2x)} \xrightarrow{\text{152}} du = \frac{-1 \cdot (-\sin(2x) \cdot 2 dx)}{\cos^2(2x)}$$

$$dv = \frac{1}{\cos^2(2x)} dx \xrightarrow{\text{167}} v = \frac{\tan(2x)}{2}$$

$$= \frac{1}{\cos(2x)} \cdot \frac{\tan(2x)}{2} - \int \frac{\tan(2x)}{2} \cdot \frac{2 \sin(2x)}{\cos^2(2x)} dx$$

$$= \frac{\sin(2x)}{2 \cos^2(2x)} - \int \frac{\sin^2(2x)}{\cos^3(2x)} dx \quad \xrightarrow{1 - \cos^2(2x)}$$

$$= \frac{\sin(2x)}{2 \cos^2(2x)} - \underbrace{\int \frac{1}{\cos^3(2x)} dx}_{I} + \int \frac{1}{\cos(2x)} dx$$

f3 > ↓

$$\frac{\ln \left| \tan \frac{2x}{2} + \frac{\pi}{4} \right|}{2}$$

$$I = \frac{\sin(2x)}{2 \cos^2(2x)} - I + \frac{\ln \left| \tan \frac{2x}{2} + \frac{\pi}{4} \right|}{2}$$

$$2I = \frac{\sin(2x)}{2 \cos^2(2x)} + \frac{\ln \left| \tan \frac{2x}{2} + \frac{\pi}{4} \right|}{2}$$

$$I = \frac{\sin(2x)}{2 \cos^2(2x)} + \frac{\ln \left| \tan \frac{2x}{2} + \frac{\pi}{4} \right|}{2} + C$$

$$\int \frac{x^2 dx}{\sqrt{4-x^2}}$$

① ③

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A 15

$$x = 2 \sin t \rightarrow$$

$$dx = 2 \cos t dt$$

$$\frac{x}{2} = \sin t$$

$$t = \arcsin \left(\frac{x}{2} \right)$$

$$\int \frac{(2 \sin t)^2 \cdot 2 \cos t \, dt}{\sqrt{4 - (2 \sin t)^2}}$$

$$= \int \frac{8 \cdot \sin^2 t \cdot \cos t}{\sqrt{4 - 4 \sin^2 t}} \, dt$$

$$\swarrow$$

$$\cos^2 t + \sin^2 t = 1$$

$$\int \frac{8 \cdot \sin^2 t \cdot \cos t}{\sqrt{\cos^2 t}} \, dt$$

$$= 8 \int \sin^2 t \, dt$$

$$8 \int \frac{1 - \cos(2t)}{2} \, dt$$

$$g \left[\frac{1}{g} t - \frac{\sin(gt)}{g \cdot g} \right] = c$$

$$t = \arcsin\left(\frac{\lambda}{g}\right) \quad \text{!!! > 3rd}$$

$$t = \arctan\left(\frac{x}{g}\right)$$

$$\frac{x}{g} = \tan t \rightarrow$$

$$\int \frac{dx}{x \sqrt{k+x^2}}$$

$$x = g \tan t$$

$$dx = \frac{g}{\cos^2 t} dt$$

(1)

$$\int \frac{\frac{g}{\cos^2 t} dt}{g \tan t \cdot \sqrt{k + k t^2 \tan^2 t}}$$



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$$1 + \tan^2 x = \frac{1}{\cos^2 x}$$

$$\int \frac{2}{\cos^2 t} \cdot 2 \cdot \tan t \cdot \sqrt{4 \cdot \frac{1}{\cos^2 t}} dt$$

$$= \int \frac{2}{\cos^2 t} \cdot 2 \cdot \tan t \cdot 2 \cdot \frac{1}{\cos t} dt$$

$$\int \frac{2}{\cancel{\cos^2 t}} \cdot 4 \cdot \frac{\sin t}{\cancel{\cos t}} \cdot \frac{1}{\cancel{\cos t}} dt$$

$$= \int \frac{1}{2} - \frac{1}{\sin t} dt$$

$$= \frac{1}{2} \ln \left| \tan \frac{t}{2} \right| + C$$

↖ limmer

$$t = \arctan\left(\frac{x}{2}\right)$$

(t)

$$\int \frac{dx}{x^3 \sqrt{x^2 - 36}}$$

$$x = \frac{6}{\cos t}$$

$$dx = \frac{-6 \cdot (-\sin t) dt}{\cos^2 t}$$

$$= \int \frac{\frac{6 \sin t}{\cos^2 t} dt}{\left(\frac{6}{\cos t}\right)^3 \cdot \sqrt{\left(\frac{6}{\cos t}\right)^2 - 36}}$$

$$\int \frac{\frac{6 \sin t}{\cos^2 t}}{\frac{6^3}{\cos^3 t} \cdot \sqrt{\frac{36}{\cos^2 t} - 36}} df$$

$$\int \frac{\frac{6 \sin t}{\cos^2 t}}{\frac{6^3}{\cos^3 t} \sqrt{\frac{36 - 36 \cos^2 t}{\cos^2 t}}}$$



$$1 - \cos^2 x = \sin^2 x$$

$$\int \frac{6 \sin t}{\cos^2 t} dt$$

$$\frac{6^3}{\cos^3 t} \quad \sqrt{\frac{36 \sin^2 t}{\cos^2 t}} \quad \downarrow$$

$$\int \frac{6 \sin t}{\cos^2 t} dt$$

$$\frac{6^3}{\cos^3 t} \cdot \frac{6 \cdot \sin t}{\cos t}$$

$$\int \left(\frac{6 \cancel{\sin t}}{\cos^2 t} + \frac{6^4 \cancel{\sin t}}{\cos^4 t} \right) dt$$

$$= \int \frac{6 \cos^4 t}{6^4 \cdot \cos^2 t} dt =$$

$$\frac{1}{6^3} \int \cos^2 t \, dt$$

$$\frac{1}{6^3} \int \frac{1 + \cos 2t}{2} \, dt$$

$$\frac{1}{6^3} \left[\frac{1}{2} t + \frac{\sin(2t)}{2 \cdot 2} \right] + C$$

$$x = \frac{6}{\cos t} \rightarrow \cos t = \frac{6}{x}$$

$$t = \arccos\left(\frac{6}{x}\right)$$

$$\int_1^2 x \, dx = \frac{x^2}{2} \Big|_1^2 = \frac{2^2}{2} - \frac{1^2}{2}$$

1.1.1?

! '310' C22W

$$\left(\int_{\text{noor}}^{x^2} t^3 \, dt \right)' = (x^2)^3 \cdot 2x$$

$$\left(\int_{x^3}^{\text{noor}} e^t \, dt \right)' = - e^{x^3} \cdot 3x^2$$

$$\left(\int_{x^2}^{x^3} \sin t \, dt \right)' =$$

$$\sin(x^3) \cdot 3x^2 - \sin(x^2) \cdot 2x$$

$$\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sin(t^2) dt}{x^6} = \left\{ \frac{0}{0} \right\} \quad \text{L'Hôpital's Rule}$$

$$= \lim_{x \rightarrow 0} \frac{\sin(x^2)^2 \cdot 2x}{6x^5} =$$

$$\lim_{x \rightarrow 0} \frac{1}{3} \cdot \frac{\sin(x^4)}{x^4} = \frac{1}{3}$$

! 201

$$\lim_{x \rightarrow \infty} \frac{x^4}{\int_0^{x^2} \ln(1+3t) dt} = \left\{ \frac{0}{0} \right\} = \text{L'Hôpital}$$

$$= \lim_{x \rightarrow \infty} \frac{2x^3}{\ln(1+3x^2) \cdot 2x}$$

$$\lim_{x \rightarrow 0} \frac{2x^2}{\ln(1+3x^2)} = \left\{ \frac{0}{0} \right\}$$

$$\lim_{x \rightarrow \infty} \left(\frac{\frac{4x}{1+3x^2}}{1+3x^2} \right)$$

$$\lim_{x \rightarrow 0} \frac{4x(1+3x^2)}{6x} = \frac{4}{6}$$

הצגת נגזרת

נתונה פונקציה $f(x)$ ערכי x קרובים לאפס

$$f(x) = \int_{-3}^{x^3+1} t(t-1)^5 dt$$

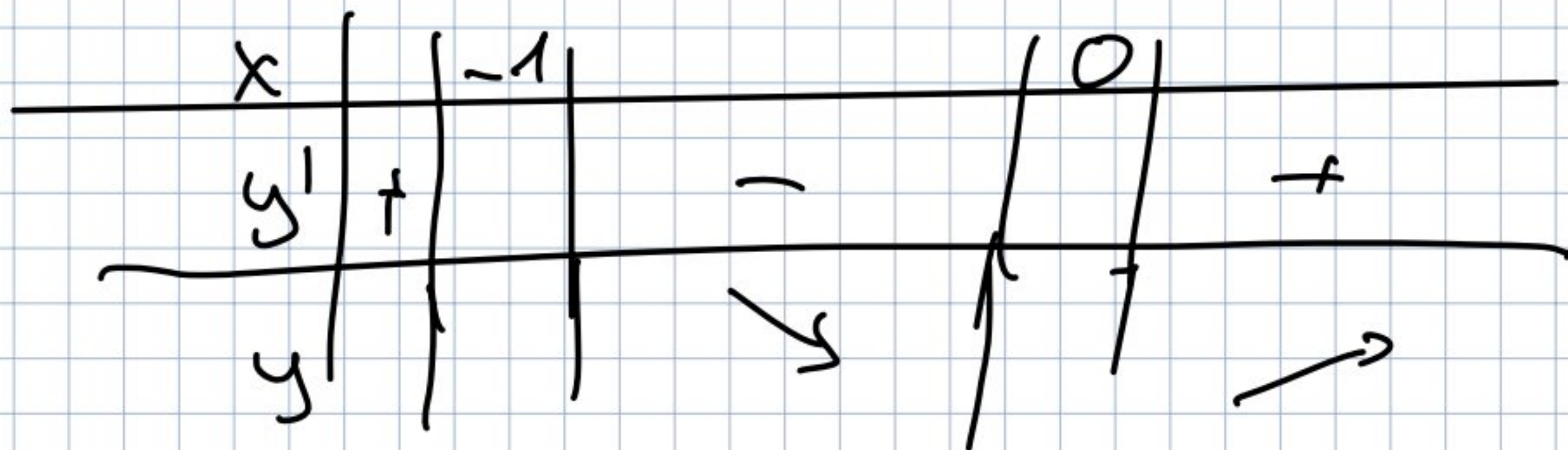
ע' ק' נ' נ' .

$$f'(x) = (x^3+1)(x^3+1-1)^5 \cdot (3x^2) = 0$$

$$(x^3+1) x^{15} \cdot 3x^2 = 0$$

$$\downarrow \\ x = -1$$

$$\searrow \\ x = 0$$



$$x = -1$$

$$x = 0$$

10, 11, 10 83